

Lesson 11: Quantum Kernel Methods

Fidelity, feature maps, precomputed kernels, and classical SVM

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Learning Outcomes

- Explain what a kernel measures conceptually.
- Understand how a quantum circuit can define a similarity function.
- Describe the hybrid quantum-kernel + classical SVM pipeline.
- Interpret the recorded 60% accuracy.

A kernel measures similarity in a feature space.

- Classical SVMs can use kernels instead of raw coordinates.
- A quantum feature map sends data into a quantum state space.
- The kernel is derived from overlap/fidelity between encoded states.

Why the Two-Moons Dataset?

- The two-moons dataset contains two curved classes.
- It is useful because a straight-line boundary is not ideal.
- The notebook standardizes the features before quantum encoding.

Custom Fidelity Function

$$F(\psi, \phi) = |\langle \psi | \phi \rangle|^2$$

- The notebook combines one circuit with the inverse of another.
- The probability of the all-zero outcome is used as a fidelity-like quantity in the custom function.
- That value becomes a similarity score between two inputs.

Quantum Kernel + Classical SVM

```
X, y = make_moons(n_samples=100, noise=0.1, random_state=42)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42)
feature_map = ZZFeatureMap(feature_dimension=2, reps=2, entanglement='linear')
sampler = StatevectorSampler()
quantum_kernel = CustomFidelityQuantumKernel(feature_map=feature_map, sampler=sampler)
X_train_kernel = quantum_kernel.evaluate(X_train)
X_test_kernel = quantum_kernel.evaluate(X_test, X_train)
svm = SVC(kernel='precomputed')
svm.fit(X_train_kernel, y_train)
y_pred = svm.predict(X_test_kernel)
accuracy = np.mean(y_pred == y_test)
print(f"Quantum Kernel SVM Accuracy: {accuracy * 100:.2f}%")
```

- `evaluate(X_train)` produces the training kernel matrix.
- `evaluate(X_test, X_train)` produces test-vs-training similarities.
- `SVC(kernel="precomputed")` receives the quantum kernel instead of raw features.
- The final prediction is still made by a classical SVM.

Quantum Kernel SVM Accuracy: 60.00%

- The recorded classifier correctly predicts 60% of the held-out examples.
- This is a demonstration of the quantum-kernel workflow, not evidence of superiority over all classical kernels.
- Kernel quality, dataset size, encoding choice and noise all matter.

Beginner Activity

- Visualize the two-moons dataset before training.
- Change the noise parameter.
- Discuss how a curved decision boundary can motivate nonlinear feature spaces.

- A kernel measures similarity.
- Feature maps define how classical inputs become quantum states.
- An SVM can consume a precomputed quantum kernel matrix.
- The notebook records 60% accuracy.

Quantum kernels use quantum state similarity as a feature-space tool for a classical learner.

Quick Check

- ❶ What does a kernel represent?
- ❷ Why use a feature map?
- ❸ Which part of the final classifier is classical?